

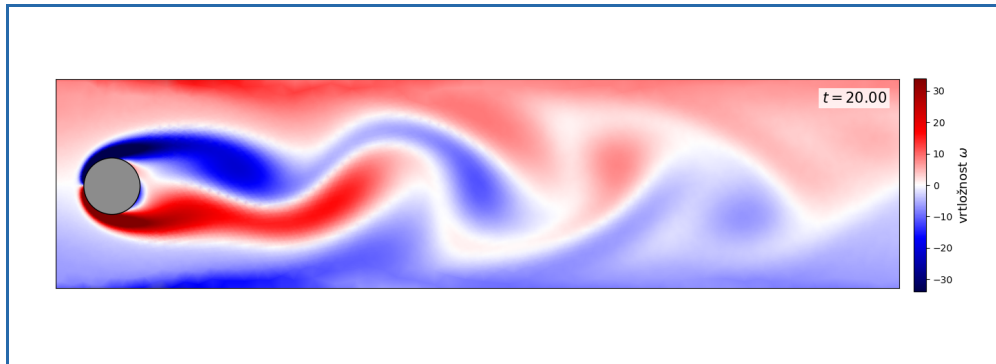
Diskretna empirijska interpolacijska metoda (DEIM)

Matko Petričić

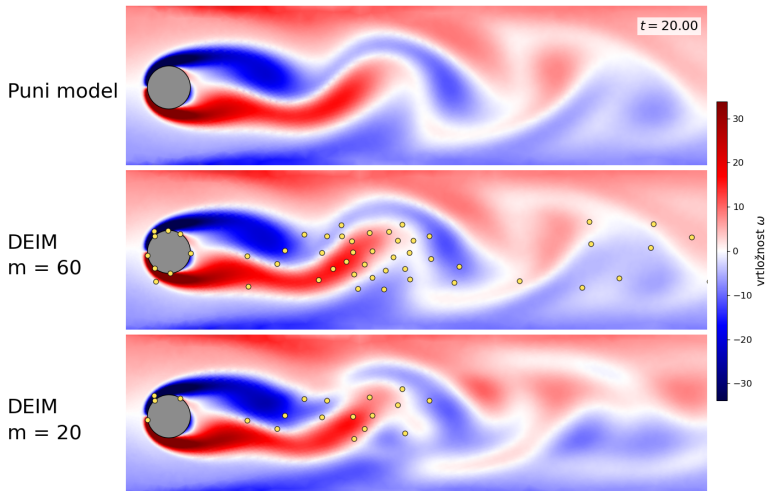
Sveučilište u Zagrebu
Prirodoslovno-matematički fakultet
Matematički odsjek

rujan 2026.

Strujanje iza valjka: puni model



Isto strujanje iz rijetkih mjerenja



POD: od matrice uzoraka do reducirane baze

$$f : \mathcal{P} \rightarrow \mathbb{R}^n, \quad Y = [f(p_1) \quad \cdots \quad f(p_{N_s})] \in \mathbb{R}^{n \times N_s}$$

$$\begin{bmatrix} \bullet & \cdots & \bullet \\ \vdots & \ddots & \vdots \\ \bullet & \cdots & \bullet \end{bmatrix} \underset{Y}{\approx} \begin{bmatrix} \bullet & \bullet \\ \vdots & \vdots \\ \bullet & \bullet \end{bmatrix} \begin{bmatrix} \bullet & \cdots & \bullet \\ \bullet & \cdots & \bullet \end{bmatrix} \underset{U_m}{U_m^T Y}$$

$$U_m \in \underset{\substack{Z \in \mathbb{R}^{n \times m} \\ Z^T Z = I_m}}{\operatorname{argmin}} \|Y - ZZ^T Y\|_2$$

$$\text{SVD} \quad \begin{cases} Y = W \Sigma V^T, & U_m = W(:, 1:m), \\ \|Y - U_m(U_m^T Y)\|_2 = \sigma_{m+1}. \end{cases}$$

$$f_m(p) = U_m c(p), \quad c(p) = U_m^T f(p)$$

Za koeficijente još trebamo svih n komponenti $f(p)$.

DEIM: Određivanje indeksa vektora iz kojih se aproksimira vektor u cijelosti

$$\left. \begin{aligned} f_m(p) = U_m c(p) \end{aligned} \right\} \begin{aligned} &\implies c(p) = U_m^T f(p), \\ &\xRightarrow{\text{DEIM}} c(p) = \dots \end{aligned}$$

$\text{rang}(U_m) = m \implies$ postoji m linearno nezavisnih redaka matrice U_m ,
pa postoji interpolacijska matrica P
za koju je $P^T U_m$ regularna.

$$P^T f(p) = P^T U_m c(p)$$

$$c(p) = \left(P^T U_m \right)^{-1} P^T f(p)$$

DEIM: Određivanje indeksa vektora iz kojih se aproksimira vektor u cijelosti

Ulaz: $U_m = [u_1 \cdots u_m] \in \mathbb{R}^{n \times m}$, **izlaz:** $\wp = (\wp_1, \dots, \wp_m)^T$.

Inicijalizacija

$$\wp_1 = \operatorname{argmax}_{1 \leq i \leq n} |u_1(i)|, \quad U = [u_1], \quad P = [e_{\wp_1}].$$

Za $\ell = 2, \dots, m$:

$$\begin{aligned} (P^T U) c &= P^T u_\ell, & r &= u_\ell - U c, \\ \wp_\ell &= \operatorname{argmax}_{1 \leq i \leq n} |r(i)|, & U &\leftarrow [U \ u_\ell], & P &\leftarrow [P \ e_{\wp_\ell}]. \end{aligned}$$

Ocjena greške = koliko je dobar potprostor $\times$ koliko su dobro odabrani indksi

Za svaki dopušteni odabir P :

$$\left\| f(p) - \hat{f}(p) \right\|_2 \leq \underbrace{\left\| \left(P^T U_m \right)^{-1} \right\|_2}_{\text{odabir indeksa}} \underbrace{\left\| \left(I - U_m U_m^T \right) f(p) \right\|_2}_{\text{POD aproksimacija}}$$

Cilj je odabrati P tako da $P^T U_m$ ima malu normu inverza.

To nas vodi do inačice DEIM-a nazvane **QDEIM**.

QDEIM koristi QR faktorizaciju s pivotiranjem

$$U_m^T \Pi = Q \begin{bmatrix} R_{11} & R_{12} \end{bmatrix}, \quad P = \Pi(:, 1:m), \quad U_m^T P = QR_{11}$$
$$\sigma_i(P^T U_m) = \sigma_i(U_m^T P) = \sigma_i(R_{11}), \quad \left\| (P^T U_m)^{-1} \right\|_2 = \frac{1}{\sigma_{\min}(R_{11})}$$

$$\prod_{k=1}^m r_{kk} = \prod_{k=1}^m \sigma_k(R_{11}) \quad (= \det(R_{11}))$$

U k -tom koraku CPQR odabire stupac s najvećim rezidualom:

$$a_j = U_m^T e_j = (U_m)_{j,:}^T, \quad \rho_j^{(k-1)} = \left\| \left(I - Q_{k-1} Q_{k-1}^T \right) a_j \right\|_2,$$

$$\wp_k = \operatorname{argmax}_{j \notin S_{k-1}} \rho_j^{(k-1)}, \quad r_{kk} = \rho_{\wp_k}^{(k-1)}.$$

Galerkinova projekcija sustava ODJ-a

$$\mathbf{M}\dot{x}(t; \mu) = f(x(t; \mu), t; \mu), \quad x(t; \mu) \in \mathbb{R}^n.$$

$$x(t; \mu) \approx x_r(t; \mu) = x_{\text{ref}} + \underbrace{\mathbf{V}_r}_{\substack{n \times r \\ r \ll n}} \underbrace{\hat{x}(t; \mu)}_{\mathbb{R}^r}, \quad \mathbf{V}_r^T \mathbf{V}_r = I_r$$

$$x_r(t; \mu) \in x_{\text{ref}} + \text{Im}(\mathbf{V}_r).$$

Stupci matrice $\mathbf{V}_r$ određuju dopuštene smjerove.

$\hat{x}(t; \mu) \in \mathbb{R}^r$: koordinate u toj bazi.

$$\mathbf{V}_r^T \mathbf{M} \mathbf{V}_r \dot{\hat{x}} = \mathbf{V}_r^T f(x_{\text{ref}} + \mathbf{V}_r \hat{x}, t; \mu)$$

DEIM interpolacijska projekcija

$$\mathbf{V}_r^T \mathbf{M} \mathbf{V}_r \dot{\hat{\mathbf{x}}} = \underbrace{\mathbf{V}_r^T f(x_r, t; \mu)}_{\in \mathbb{R}^n}.$$

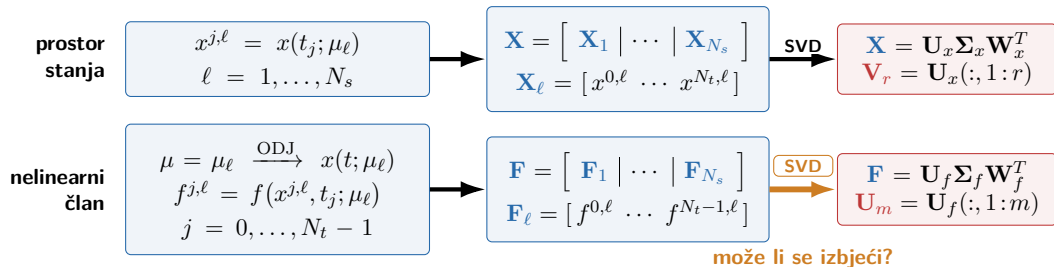
DEIM aproksimacija

$$f(x_r, t; \mu) \approx \mathbf{U}_m \left(\mathbf{P}^T \mathbf{U}_m \right)^{-1} \underbrace{\mathbf{P}^T f(x_r, t; \mu)}_{\substack{m \text{ odabranih} \\ \text{komponenti}}}, \quad m \ll n.$$

$$\mathbf{V}_r^T \mathbf{M} \mathbf{V}_r \dot{\hat{\mathbf{x}}} = \underbrace{\mathbf{V}_r^T \mathbf{U}_m \left(\mathbf{P}^T \mathbf{U}_m \right)^{-1}}_{\substack{\text{računa se u} \\ \text{offline fazi}}} \underbrace{\mathbf{P}^T f(x_r, t; \mu)}_{\substack{\text{evaluira se u} \\ \text{online fazi}}}.$$

Uz izravnu evaluaciju u *online* fazi računamo m , umjesto n , komponenti funkcije.

Standardni POD–DEIM koristi dva SVD-a



EksPLICITNA Eulerova metoda

$$\mathbf{M}(x^k - x^{k-1}) = \Delta t f^{k-1} \implies f^{k-1} = \frac{1}{\Delta t} (\mathbf{M}x^k - \mathbf{M}x^{k-1})$$

$$\text{Im}(\mathbf{F}) \subseteq \text{Im}(\mathbf{M}\mathbf{X})$$

SNS: dva izbora baze nelinearnog člana

$$\mathbf{M}\dot{\mathbf{x}}_r = \mathbf{M}\mathbf{V}_r\dot{\hat{\mathbf{x}}} \in \text{Im}(\mathbf{M}\mathbf{V}_r), \quad f(x_r, t; \mu) \approx \tilde{f}_m(x_r, t; \mu) \in \text{Im}(\mathbf{B}_m)$$

Uvjet kompatibilnosti prostora

$$\text{Im}(\mathbf{M}\mathbf{V}_r) \subseteq \text{Im}(\mathbf{B}_m)$$

1. Osnovni izbor

$$m = r, \quad \mathbf{B}_m = \mathbf{M}\mathbf{V}_r$$

$$\text{Im}(\mathbf{B}_m) = \text{Im}(\mathbf{M}\mathbf{V}_r)$$

Najjeftiniji izbor: prostor desne strane i model stanja imaju dimenziju r .

2. Prošireni izbor

$$\mathbf{V}_m = [\mathbf{V}_r \quad \mathbf{V}_E], \quad m > r,$$

$$\mathbf{B}_m = \mathbf{M}\mathbf{V}_m,$$

$$\text{Im}(\mathbf{M}\mathbf{V}_r) \subseteq \text{Im}(\mathbf{B}_m)$$

Model stanja ostaje dimenzije r .

$\mathbf{V}_E$: $m - r$ dodatnih POD vektora.

Ortonormiranje ekonomskom QR faktorizacijom

$$\mathbf{B}_m = \mathbf{U}_m \mathbf{R}, \quad \mathbf{U}_m^T \mathbf{U}_m = \mathbf{I}_m, \quad \text{Im}(\mathbf{U}_m) = \text{Im}(\mathbf{B}_m).$$

Navier–Stokes: puni model i skup podataka

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} - \nu \Delta \mathbf{u} + \nabla p = 0, \quad \nabla \cdot \mathbf{u} = 0, \quad \nu = 10^{-3}.$$

Parabolična ulazna brzina

$$u_{\text{in},1}(y, t; \mu) = a(t; \mu) \frac{4y(H - y)}{H^2}$$

$a(t; \mu)$: maksimalna ulazna brzina

$$a(t; \mu) = \mu \sin\left(\frac{\pi}{2} \min\left\{\frac{t}{4}, 1\right\}\right)$$

$$\mu = U_{\text{max}} \in [0.7, 1.0]$$

$H = 0.41$: visina kanala, $0 \leq y \leq H$

Diskretizacija i uzorci

dimenzija polja $n = 11\,994$

vremenski interval $0 \leq t \leq 8$

vremenski korak $\Delta t = 0.01$

parametri $N_\mu = 31$

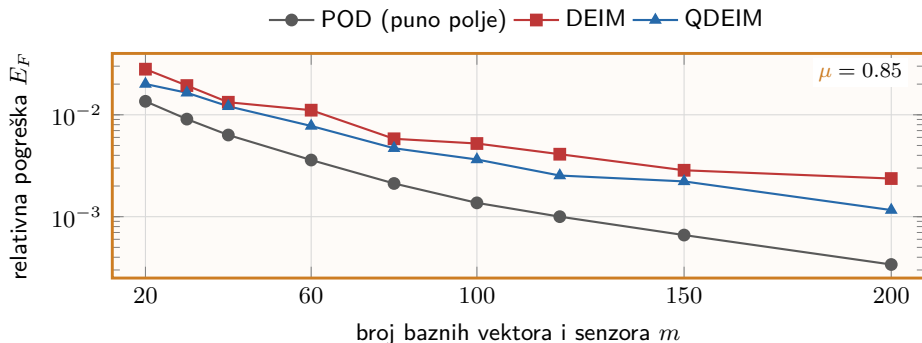
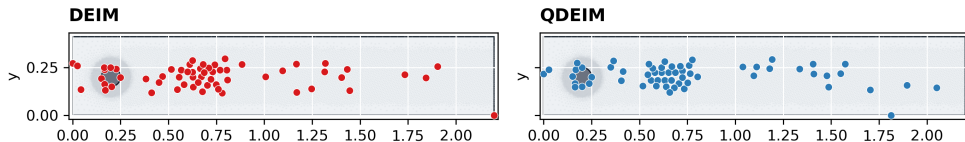
uzorci po parametru $N_t = 400$

ukupno uzoraka $N = 12\,400$

$$\mu_i = 0.70, 0.71, \dots, 1.00$$

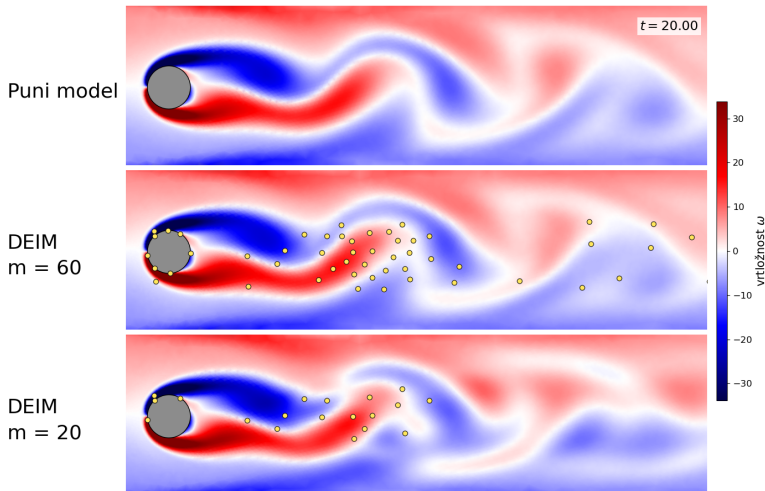
$$\text{Rekonstruiramo vrtložnost } \omega = \frac{\partial u_2}{\partial x} - \frac{\partial u_1}{\partial y}.$$

Navier–Stokes: rekonstrukcija iz rijetkih mjerenja



Isto strujanje iz rijetkih mjerenja

$$\mu = 0.855, \quad 0 \leq t \leq 30$$



Hvala na pažnji!